

# US Patent & Trademark Office

## Patent Public Search | Text View

---

|                      |                      |
|----------------------|----------------------|
| United States Patent | 12396752             |
| Kind Code            | B2                   |
| Date of Patent       | August 26, 2025      |
| Inventor(s)          | Byager; Rune Frimand |

---

### Inserter system with transport protection

---

#### Abstract

Single use inserter system comprising a housing part and a lid part, the inserter system comprising a carrier part in an initial position in the housing and a drive unit comprising a first drive part attached to the housing, the drive unit comprising at least one spring element supported by the first drive part, wherein the drive unit is configured for moving the carrier part from a first position to a second position in relation to the housing in an insertion direction along a first axis. The inserter system comprises a transcutaneous device having a proximal surface and a distal surface, wherein the transcutaneous device is detachably attached to the carrier part, the inserter system comprising at least one transport protection element preventing movement of the carrier part in a direction along the first axis thereby supporting the carrier part in the initial position.

---

|                   |                                                 |
|-------------------|-------------------------------------------------|
| <b>Inventors:</b> | <b>Byager; Rune Frimand (Frederiksberg, DK)</b> |
| <b>Applicant:</b> | <b>UnoMedical A/S (Birkerød, DK)</b>            |
| <b>Family ID:</b> | <b>1000008780634</b>                            |
| <b>Assignee:</b>  | <b>Unomedical A/S (Birkerød, DK)</b>            |
| <b>Appl. No.:</b> | <b>17/667978</b>                                |
| <b>Filed:</b>     | <b>February 09, 2022</b>                        |

#### Prior Publication Data

|                            |                         |
|----------------------------|-------------------------|
| <b>Document Identifier</b> | <b>Publication Date</b> |
| US 20220160396 A1          | May. 26, 2022           |

#### Foreign Application Priority Data

|    |          |               |
|----|----------|---------------|
| EP | 11158118 | Mar. 14, 2011 |
|----|----------|---------------|

#### Related U.S. Application Data

continuation parent-doc US 16265832 20190201 US 11317944 child-doc US 17667978  
continuation parent-doc US 14003193 US 10194938 20190205 WO PCT/EP2012/053722  
20120305 child-doc US 16265832  
us-provisional-application US 61452836 20110315

---

## Publication Classification

**Int. Cl.:** **A61B17/34** (20060101); **A61B5/00** (20060101); **A61M5/158** (20060101); A61B17/00 (20060101)

**U.S. Cl.:**

**CPC** **A61B17/3468** (20130101); **A61B5/686** (20130101); **A61B17/3415** (20130101); **A61M5/158** (20130101); A61B2017/0023 (20130101); A61B2017/00367 (20130101); A61B2017/347 (20130101); A61M2005/1585 (20130101)

## Field of Classification Search

**CPC:** A61M (2005/1585); A61M (25/0612); A61M (25/0618); A61M (25/0625); A61B (17/3468); A61B (5/6848); A61B (5/151-15198); A61B (5/150412); A61B (5/150419); A61B (5/150427); A61B (5/150503); A61B (5/1519); A61B (5/15101); A61B (5/150541); A61B (5/150549); A61B (5/150557); A61B (5/150564); A61B (5/150908); A61B (5/150717)

---

## References Cited

### U.S. PATENT DOCUMENTS

| Patent No. | Issued Date | Patentee Name      | U.S. Cl. | CPC              |
|------------|-------------|--------------------|----------|------------------|
| 5628765    | 12/1996     | Morita             | 606/181  | A61B<br>5/150312 |
| 5964718    | 12/1998     | Duchon et al.      | N/A      | N/A              |
| 6390990    | 12/2001     | Marshall           | 600/573  | A61B<br>5/150519 |
| 10071210   | 12/2017     | Gray               | N/A      | N/A              |
| 10292641   | 12/2018     | Bureau et al.      | N/A      | N/A              |
| 10293101   | 12/2018     | Brewer et al.      | N/A      | N/A              |
| 10369274   | 12/2018     | O'Connor et al.    | N/A      | N/A              |
| 10369289   | 12/2018     | Cabiri et al.      | N/A      | N/A              |
| 10376638   | 12/2018     | Levesque et al.    | N/A      | N/A              |
| 10413661   | 12/2018     | Kamen et al.       | N/A      | N/A              |
| 10432403   | 12/2018     | Moskal             | N/A      | N/A              |
| 10434245   | 12/2018     | Yodfat et al.      | N/A      | N/A              |
| 10434247   | 12/2018     | Cole et al.        | N/A      | N/A              |
| 10434248   | 12/2018     | Penake et al.      | N/A      | N/A              |
| 10434253   | 12/2018     | DiPerna et al.     | N/A      | N/A              |
| 10434285   | 12/2018     | Schoonmaker et al. | N/A      | N/A              |
| 10438696   | 12/2018     | Shapley et al.     | N/A      | N/A              |
| 10441356   | 12/2018     | Zarins et al.      | N/A      | N/A              |

|          |         |                     |     |     |
|----------|---------|---------------------|-----|-----|
| 10441713 | 12/2018 | Feldman et al.      | N/A | N/A |
| 10441718 | 12/2018 | Tchao et al.        | N/A | N/A |
| 10441723 | 12/2018 | Nazzaro             | N/A | N/A |
| 10441775 | 12/2018 | Schriver et al.     | N/A | N/A |
| 10449290 | 12/2018 | Shapley et al.      | N/A | N/A |
| 10449291 | 12/2018 | Hadian et al.       | N/A | N/A |
| 10449296 | 12/2018 | Kapas et al.        | N/A | N/A |
| 10449306 | 12/2018 | Grover et al.       | N/A | N/A |
| 10463572 | 12/2018 | Shor et al.         | N/A | N/A |
| 10463785 | 12/2018 | Dewey               | N/A | N/A |
| 10463787 | 12/2018 | Shor et al.         | N/A | N/A |
| 10463791 | 12/2018 | Shergold et al.     | N/A | N/A |
| 10471203 | 12/2018 | Chappel et al.      | N/A | N/A |
| 10471206 | 12/2018 | Dittrich            | N/A | N/A |
| 10478550 | 12/2018 | Hadvary et al.      | N/A | N/A |
| 10478552 | 12/2018 | Cronenberg et al.   | N/A | N/A |
| 10478554 | 12/2018 | Bazargan et al.     | N/A | N/A |
| 10478555 | 12/2018 | Radojicic           | N/A | N/A |
| 10481024 | 12/2018 | Wade et al.         | N/A | N/A |
| 10483000 | 12/2018 | Saint et al.        | N/A | N/A |
| 10485923 | 12/2018 | Schiendzielorz      | N/A | N/A |
| 10485926 | 12/2018 | Vanderveen et al.   | N/A | N/A |
| 10485937 | 12/2018 | Yodfat et al.       | N/A | N/A |
| 10489617 | 12/2018 | Salem et al.        | N/A | N/A |
| 10493201 | 12/2018 | Cole et al.         | N/A | N/A |
| 10493202 | 12/2018 | Hayter              | N/A | N/A |
| 10493203 | 12/2018 | Yodfat et al.       | N/A | N/A |
| 10500352 | 12/2018 | Grant et al.        | N/A | N/A |
| 10507316 | 12/2018 | Fielder et al.      | N/A | N/A |
| 10512724 | 12/2018 | Renstad et al.      | N/A | N/A |
| 10525193 | 12/2019 | Schauderna          | N/A | N/A |
| 10525247 | 12/2019 | Bellrichard et al.  | N/A | N/A |
| 10532150 | 12/2019 | Bazargan et al.     | N/A | N/A |
| 10532151 | 12/2019 | Wei                 | N/A | N/A |
| 10532155 | 12/2019 | Schiendzielorz      | N/A | N/A |
| 10532159 | 12/2019 | Torsten et al.      | N/A | N/A |
| 10532835 | 12/2019 | Chong et al.        | N/A | N/A |
| 10537681 | 12/2019 | Tan-Malecki et al.  | N/A | N/A |
| 10539481 | 12/2019 | Plahey et al.       | N/A | N/A |
| 10542921 | 12/2019 | Kuhn                | N/A | N/A |
| 10542936 | 12/2019 | Goldberg et al.     | N/A | N/A |
| 10549029 | 12/2019 | Agard et al.        | N/A | N/A |
| 10549033 | 12/2019 | Shimizu             | N/A | N/A |
| 10549034 | 12/2019 | Eggert et al.       | N/A | N/A |
| 10549036 | 12/2019 | Starkweather et al. | N/A | N/A |
| 10549079 | 12/2019 | Burton et al.       | N/A | N/A |
| 10556059 | 12/2019 | Cross et al.        | N/A | N/A |
| 10556063 | 12/2019 | Murphy, Jr. et al.  | N/A | N/A |
| 10561785 | 12/2019 | Roy et al.          | N/A | N/A |
| 10561788 | 12/2019 | Roy                 | N/A | N/A |

|          |         |                       |     |     |
|----------|---------|-----------------------|-----|-----|
| 10561789 | 12/2019 | Mastrototaro et al.   | N/A | N/A |
| 10561826 | 12/2019 | Amano et al.          | N/A | N/A |
| 10561831 | 12/2019 | Kato                  | N/A | N/A |
| 10569011 | 12/2019 | Dilanni et al.        | N/A | N/A |
| 10569012 | 12/2019 | Schabbach et al.      | N/A | N/A |
| 10569014 | 12/2019 | Hanson et al.         | N/A | N/A |
| 10576199 | 12/2019 | Sealfon et al.        | N/A | N/A |
| 10576203 | 12/2019 | Amon et al.           | N/A | N/A |
| 10576204 | 12/2019 | Estes et al.          | N/A | N/A |
| 10583241 | 12/2019 | Wu et al.             | N/A | N/A |
| 10583247 | 12/2019 | Mandro                | N/A | N/A |
| 10589023 | 12/2019 | Cindrich et al.       | N/A | N/A |
| 10589028 | 12/2019 | Cabiri et al.         | N/A | N/A |
| 10596317 | 12/2019 | Nakanishi             | N/A | N/A |
| 10596362 | 12/2019 | Fielder et al.        | N/A | N/A |
| 10610638 | 12/2019 | Cabiri et al.         | N/A | N/A |
| 10610639 | 12/2019 | Cabiri et al.         | N/A | N/A |
| 10610644 | 12/2019 | Mazlish et al.        | N/A | N/A |
| 10617817 | 12/2019 | Hwang et al.          | N/A | N/A |
| 10617820 | 12/2019 | O'Connor et al.       | N/A | N/A |
| 10625016 | 12/2019 | Amon et al.           | N/A | N/A |
| 10625017 | 12/2019 | Searle et al.         | N/A | N/A |
| 10625018 | 12/2019 | Destefano et al.      | N/A | N/A |
| 10632248 | 12/2019 | Stefanov et al.       | N/A | N/A |
| 10632249 | 12/2019 | Marbet et al.         | N/A | N/A |
| 10632253 | 12/2019 | Uchiyama et al.       | N/A | N/A |
| 10632256 | 12/2019 | Sasaki                | N/A | N/A |
| 10632257 | 12/2019 | Estes et al.          | N/A | N/A |
| 10635784 | 12/2019 | Rubalcaba, Jr. et al. | N/A | N/A |
| 10639417 | 12/2019 | Roberts               | N/A | N/A |
| 10639418 | 12/2019 | Kamen et al.          | N/A | N/A |
| 10639661 | 12/2019 | Fontana               | N/A | N/A |
| 10646643 | 12/2019 | Cabiri et al.         | N/A | N/A |
| 10646652 | 12/2019 | McCullough et al.     | N/A | N/A |
| 10646653 | 12/2019 | Despa et al.          | N/A | N/A |
| 10653828 | 12/2019 | Brown et al.          | N/A | N/A |
| 10653829 | 12/2019 | Barchen et al.        | N/A | N/A |
| 10653833 | 12/2019 | Kamen et al.          | N/A | N/A |
| 10653835 | 12/2019 | Dobbles et al.        | N/A | N/A |
| 10653846 | 12/2019 | Weibel et al.         | N/A | N/A |
| 10656894 | 12/2019 | Fryman                | N/A | N/A |
| 10661006 | 12/2019 | Antonio et al.        | N/A | N/A |
| 10661007 | 12/2019 | Estes                 | N/A | N/A |
| 10661008 | 12/2019 | Brewer et al.         | N/A | N/A |
| 10661067 | 12/2019 | Kodama                | N/A | N/A |
| 10668209 | 12/2019 | Montalvo et al.       | N/A | N/A |
| 10668210 | 12/2019 | Kamen et al.          | N/A | N/A |
| 10668213 | 12/2019 | Cabiri                | N/A | N/A |
| 10668227 | 12/2019 | Caspers               | N/A | N/A |
| 10675055 | 12/2019 | Chong et al.          | N/A | N/A |

|          |         |                    |     |     |
|----------|---------|--------------------|-----|-----|
| 10675333 | 12/2019 | Ning et al.        | N/A | N/A |
| 10675404 | 12/2019 | Pizzochero et al.  | N/A | N/A |
| 10682458 | 12/2019 | Wu et al.          | N/A | N/A |
| 10682460 | 12/2019 | Adams et al.       | N/A | N/A |
| 10682461 | 12/2019 | Oakes              | N/A | N/A |
| 10682463 | 12/2019 | Kamen et al.       | N/A | N/A |
| 10685749 | 12/2019 | Hayter et al.      | N/A | N/A |
| 10688241 | 12/2019 | Yang               | N/A | N/A |
| 10688243 | 12/2019 | Cabiri             | N/A | N/A |
| 10688294 | 12/2019 | Cowan et al.       | N/A | N/A |
| 10709834 | 12/2019 | Chiu et al.        | N/A | N/A |
| 10716891 | 12/2019 | Saab et al.        | N/A | N/A |
| 10716893 | 12/2019 | Gray et al.        | N/A | N/A |
| 10716895 | 12/2019 | Brewer et al.      | N/A | N/A |
| 10716896 | 12/2019 | O'Connor et al.    | N/A | N/A |
| 10716926 | 12/2019 | Burton et al.      | N/A | N/A |
| 10719584 | 12/2019 | Drew               | N/A | N/A |
| 10722640 | 12/2019 | McLaughlin         | N/A | N/A |
| 10722643 | 12/2019 | Gray et al.        | N/A | N/A |
| 10722646 | 12/2019 | Cole et al.        | N/A | N/A |
| 10722647 | 12/2019 | Gray               | N/A | N/A |
| 10722650 | 12/2019 | Duke et al.        | N/A | N/A |
| 10722661 | 12/2019 | Mandro et al.      | N/A | N/A |
| 10729842 | 12/2019 | Hooven et al.      | N/A | N/A |
| 10729844 | 12/2019 | Cole et al.        | N/A | N/A |
| 10729849 | 12/2019 | Finan et al.       | N/A | N/A |
| 10737015 | 12/2019 | Estes              | N/A | N/A |
| 10737016 | 12/2019 | Smith et al.       | N/A | N/A |
| 10737021 | 12/2019 | Deck               | N/A | N/A |
| 10737022 | 12/2019 | Mou et al.         | N/A | N/A |
| 10737024 | 12/2019 | Schmid             | N/A | N/A |
| 10737026 | 12/2019 | Teutsch            | N/A | N/A |
| 10737038 | 12/2019 | Cole et al.        | N/A | N/A |
| 10744257 | 12/2019 | Mandro et al.      | N/A | N/A |
| 10751467 | 12/2019 | Kamen et al.       | N/A | N/A |
| 10751468 | 12/2019 | Abal               | N/A | N/A |
| 10751476 | 12/2019 | Gazeley et al.     | N/A | N/A |
| 10751478 | 12/2019 | Nazzaro            | N/A | N/A |
| 10757219 | 12/2019 | Moskal             | N/A | N/A |
| 10758675 | 12/2019 | Mazlish et al.     | N/A | N/A |
| 10758683 | 12/2019 | Gibson et al.      | N/A | N/A |
| 10758721 | 12/2019 | Sonderegger et al. | N/A | N/A |
| 10765801 | 12/2019 | McCullough         | N/A | N/A |
| 10765803 | 12/2019 | Gonnelli           | N/A | N/A |
| 10765807 | 12/2019 | Allis et al.       | N/A | N/A |
| 10772796 | 12/2019 | Kavazov            | N/A | N/A |
| 10773019 | 12/2019 | Searle et al.      | N/A | N/A |
| 10780215 | 12/2019 | Rosinko et al.     | N/A | N/A |
| 10780216 | 12/2019 | Farra              | N/A | N/A |
| 10780217 | 12/2019 | Nazzaro et al.     | N/A | N/A |

|              |         |                   |         |                  |
|--------------|---------|-------------------|---------|------------------|
| 10780220     | 12/2019 | Gray              | N/A     | N/A              |
| 10780223     | 12/2019 | Desborough et al. | N/A     | N/A              |
| 10792419     | 12/2019 | Kamen et al.      | N/A     | N/A              |
| 10792424     | 12/2019 | Sasaki            | N/A     | N/A              |
| 10792425     | 12/2019 | Joseph et al.     | N/A     | N/A              |
| 10792440     | 12/2019 | Mandro et al.     | N/A     | N/A              |
| 10799630     | 12/2019 | McCullough        | N/A     | N/A              |
| 10799631     | 12/2019 | Barmaimon et al.  | N/A     | N/A              |
| 10799632     | 12/2019 | Kohlbrecher       | N/A     | N/A              |
| 10806851     | 12/2019 | Rosinko           | N/A     | N/A              |
| 10806854     | 12/2019 | O'Connor et al.   | N/A     | N/A              |
| 10806855     | 12/2019 | Destefano et al.  | N/A     | N/A              |
| 10806859     | 12/2019 | Desborough et al. | N/A     | N/A              |
| 10814061     | 12/2019 | Bene et al.       | N/A     | N/A              |
| 10814062     | 12/2019 | Gyory             | N/A     | N/A              |
| 11135362     | 12/2020 | DiPerna et al.    | N/A     | N/A              |
| 11136971     | 12/2020 | Kamen et al.      | N/A     | N/A              |
| 11136972     | 12/2020 | Kamen et al.      | N/A     | N/A              |
| 11173244     | 12/2020 | Agard et al.      | N/A     | N/A              |
| 11179514     | 12/2020 | Yodfat et al.     | N/A     | N/A              |
| 11241534     | 12/2021 | Miller et al.     | N/A     | N/A              |
| 11261858     | 12/2021 | Kamen et al.      | N/A     | N/A              |
| 11293425     | 12/2021 | Kamen et al.      | N/A     | N/A              |
| 11339774     | 12/2021 | Gray et al.       | N/A     | N/A              |
| 11391273     | 12/2021 | Kamen et al.      | N/A     | N/A              |
| 11406753     | 12/2021 | Gray et al.       | N/A     | N/A              |
| 11408414     | 12/2021 | Kamen et al.      | N/A     | N/A              |
| 11413391     | 12/2021 | Gray              | N/A     | N/A              |
| 11491273     | 12/2021 | Gray              | N/A     | N/A              |
| 11497846     | 12/2021 | Kamen et al.      | N/A     | N/A              |
| 11534543     | 12/2021 | Kamen et al.      | N/A     | N/A              |
| 11690952     | 12/2022 | Kamen et al.      | N/A     | N/A              |
| 11690953     | 12/2022 | Lawrence et al.   | N/A     | N/A              |
| 11712513     | 12/2022 | Gray et al.       | N/A     | N/A              |
| 11712517     | 12/2022 | Lawrence et al.   | N/A     | N/A              |
| 11717609     | 12/2022 | Kamen et al.      | N/A     | N/A              |
| 11738139     | 12/2022 | Gray              | N/A     | N/A              |
| 11786651     | 12/2022 | Kamen et al.      | N/A     | N/A              |
| 11813427     | 12/2022 | Kamen et al.      | N/A     | N/A              |
| 11904134     | 12/2023 | Gray              | N/A     | N/A              |
| 11918782     | 12/2023 | Jennewine         | N/A     | N/A              |
| 11992650     | 12/2023 | Kamen et al.      | N/A     | N/A              |
| 2002/0103499 | 12/2001 | Perez et al.      | N/A     | N/A              |
| 2003/0187394 | 12/2002 | Wilkinson et al.  | N/A     | N/A              |
| 2004/0127818 | 12/2003 | Roe et al.        | N/A     | N/A              |
| 2005/0065466 | 12/2004 | Vedrine           | N/A     | N/A              |
| 2005/0101912 | 12/2004 | Faust et al.      | N/A     | N/A              |
| 2006/0106346 | 12/2005 | Sullivan et al.   | N/A     | N/A              |
| 2007/0010841 | 12/2006 | Teo               | 606/181 | A61B<br>5/150022 |

|              |         |                      |     |     |
|--------------|---------|----------------------|-----|-----|
| 2007/0021729 | 12/2006 | Mogensen et al.      | N/A | N/A |
| 2007/0124002 | 12/2006 | Estes et al.         | N/A | N/A |
| 2007/0191770 | 12/2006 | Moberg et al.        | N/A | N/A |
| 2008/0215035 | 12/2007 | Yodfat et al.        | N/A | N/A |
| 2008/0234630 | 12/2007 | Iddan et al.         | N/A | N/A |
| 2009/0218243 | 12/2008 | Gyrn et al.          | N/A | N/A |
| 2009/0326453 | 12/2008 | Adams et al.         | N/A | N/A |
| 2010/0049128 | 12/2009 | McKenzie et al.      | N/A | N/A |
| 2010/0094251 | 12/2009 | Estes                | N/A | N/A |
| 2010/0135831 | 12/2009 | Jacobsen             | N/A | N/A |
| 2010/0145303 | 12/2009 | Yodfat et al.        | N/A | N/A |
| 2010/0168670 | 12/2009 | Srisathapat et al.   | N/A | N/A |
| 2010/0241103 | 12/2009 | Kraft et al.         | N/A | N/A |
| 2011/0028982 | 12/2010 | Lacy                 | N/A | N/A |
| 2011/0040247 | 12/2010 | Mandro et al.        | N/A | N/A |
| 2011/0112484 | 12/2010 | Carter et al.        | N/A | N/A |
| 2011/0112696 | 12/2010 | Yodfat et al.        | N/A | N/A |
| 2011/0118578 | 12/2010 | Timmerman            | N/A | N/A |
| 2011/0160652 | 12/2010 | Yodfat et al.        | N/A | N/A |
| 2011/0160666 | 12/2010 | Hanson et al.        | N/A | N/A |
| 2011/0313357 | 12/2010 | Skutnik et al.       | N/A | N/A |
| 2012/0078170 | 12/2011 | Smith et al.         | N/A | N/A |
| 2012/0136300 | 12/2011 | Schoonmaker et al.   | N/A | N/A |
| 2012/0150123 | 12/2011 | Lawrence et al.      | N/A | N/A |
| 2012/0209085 | 12/2011 | Degen et al.         | N/A | N/A |
| 2012/0238851 | 12/2011 | Kamen et al.         | N/A | N/A |
| 2013/0046239 | 12/2012 | Gonnelli et al.      | N/A | N/A |
| 2013/0046508 | 12/2012 | Sur et al.           | N/A | N/A |
| 2013/0053823 | 12/2012 | Fiering et al.       | N/A | N/A |
| 2013/0060233 | 12/2012 | O'Connor et al.      | N/A | N/A |
| 2013/0138075 | 12/2012 | Lambert              | N/A | N/A |
| 2013/0226138 | 12/2012 | Sia                  | N/A | N/A |
| 2013/0237955 | 12/2012 | Neta et al.          | N/A | N/A |
| 2014/0025002 | 12/2013 | Qi et al.            | N/A | N/A |
| 2014/0031793 | 12/2013 | Constantineau et al. | N/A | N/A |
| 2014/0052096 | 12/2013 | Searle et al.        | N/A | N/A |
| 2014/0054883 | 12/2013 | Lanigan et al.       | N/A | N/A |
| 2014/0127048 | 12/2013 | Dilanni et al.       | N/A | N/A |
| 2014/0128815 | 12/2013 | Cabiri et al.        | N/A | N/A |
| 2014/0276379 | 12/2013 | Uram et al.          | N/A | N/A |
| 2014/0276536 | 12/2013 | Estes                | N/A | N/A |
| 2014/0323961 | 12/2013 | Blomquist et al.     | N/A | N/A |
| 2014/0358112 | 12/2013 | Smith et al.         | N/A | N/A |
| 2015/0025503 | 12/2014 | Searle et al.        | N/A | N/A |
| 2015/0073384 | 12/2014 | Limaye               | N/A | N/A |
| 2015/0080799 | 12/2014 | Schneider et al.     | N/A | N/A |
| 2015/0080800 | 12/2014 | Cronenberg           | N/A | N/A |
| 2015/0105720 | 12/2014 | Montalvo et al.      | N/A | N/A |
| 2015/0112269 | 12/2014 | Sumida et al.        | N/A | N/A |
| 2015/0209505 | 12/2014 | Hanson et al.        | N/A | N/A |

|              |         |                    |     |     |
|--------------|---------|--------------------|-----|-----|
| 2015/0273201 | 12/2014 | Tallarida et al.   | N/A | N/A |
| 2015/0314117 | 12/2014 | Arami et al.       | N/A | N/A |
| 2016/0051750 | 12/2015 | Tsoukalis          | N/A | N/A |
| 2016/0074578 | 12/2015 | Xu et al.          | N/A | N/A |
| 2016/0082182 | 12/2015 | Gregory et al.     | N/A | N/A |
| 2016/0089056 | 12/2015 | Limaye et al.      | N/A | N/A |
| 2016/0089524 | 12/2015 | Anderson           | N/A | N/A |
| 2016/0144105 | 12/2015 | Hooven et al.      | N/A | N/A |
| 2016/0193407 | 12/2015 | Qin et al.         | N/A | N/A |
| 2016/0346469 | 12/2015 | Shubinsky et al.   | N/A | N/A |
| 2017/0080157 | 12/2016 | Cabiri et al.      | N/A | N/A |
| 2017/0100542 | 12/2016 | Norton et al.      | N/A | N/A |
| 2017/0232191 | 12/2016 | Smith et al.       | N/A | N/A |
| 2017/0258987 | 12/2016 | Caspers            | N/A | N/A |
| 2017/0290971 | 12/2016 | Hedmann et al.     | N/A | N/A |
| 2017/0296741 | 12/2016 | Gregory            | N/A | N/A |
| 2017/0296742 | 12/2016 | Stefanov           | N/A | N/A |
| 2017/0340827 | 12/2016 | Nazzaro et al.     | N/A | N/A |
| 2017/0340841 | 12/2016 | Sasaki             | N/A | N/A |
| 2017/0351841 | 12/2016 | Moskal             | N/A | N/A |
| 2017/0351851 | 12/2016 | Wang et al.        | N/A | N/A |
| 2017/0368260 | 12/2016 | McCullough et al.  | N/A | N/A |
| 2018/0008768 | 12/2017 | Prescher et al.    | N/A | N/A |
| 2018/0028744 | 12/2017 | Kim                | N/A | N/A |
| 2018/0036476 | 12/2017 | McCullough et al.  | N/A | N/A |
| 2018/0071450 | 12/2017 | Ruhland            | N/A | N/A |
| 2018/0110420 | 12/2017 | Pekander           | N/A | N/A |
| 2018/0185573 | 12/2017 | Niklaus            | N/A | N/A |
| 2018/0193563 | 12/2017 | Krasnow et al.     | N/A | N/A |
| 2018/0200434 | 12/2017 | Mazlish et al.     | N/A | N/A |
| 2018/0200440 | 12/2017 | Mazlish et al.     | N/A | N/A |
| 2018/0207360 | 12/2017 | Jurecich et al.    | N/A | N/A |
| 2018/0214635 | 12/2017 | Raman et al.       | N/A | N/A |
| 2018/0221571 | 12/2017 | Carbone et al.     | N/A | N/A |
| 2018/0228967 | 12/2017 | Hopkins et al.     | N/A | N/A |
| 2018/0271455 | 12/2017 | Zhong et al.       | N/A | N/A |
| 2018/0280607 | 12/2017 | Richards et al.    | N/A | N/A |
| 2018/0280608 | 12/2017 | Gillett et al.     | N/A | N/A |
| 2018/0280619 | 12/2017 | Finan et al.       | N/A | N/A |
| 2018/0291882 | 12/2017 | Algawi et al.      | N/A | N/A |
| 2018/0296757 | 12/2017 | Finan et al.       | N/A | N/A |
| 2018/0344926 | 12/2017 | Brandenburg et al. | N/A | N/A |
| 2018/0361061 | 12/2017 | Andretta           | N/A | N/A |
| 2018/0372085 | 12/2017 | Veischof et al.    | N/A | N/A |
| 2019/0009019 | 12/2018 | Shor et al.        | N/A | N/A |
| 2019/0009022 | 12/2018 | Oakes              | N/A | N/A |
| 2019/0009023 | 12/2018 | Diperna et al.     | N/A | N/A |
| 2019/0015585 | 12/2018 | Smith              | N/A | N/A |
| 2019/0022317 | 12/2018 | Uddin et al.       | N/A | N/A |
| 2019/0060562 | 12/2018 | Olivas et al.      | N/A | N/A |



|              |         |                    |     |     |
|--------------|---------|--------------------|-----|-----|
| 2019/0083057 | 12/2018 | Saul et al.        | N/A | N/A |
| 2019/0083702 | 12/2018 | Nekouzadeh et al.  | N/A | N/A |
| 2019/0091404 | 12/2018 | Nazzaro et al.     | N/A | N/A |
| 2019/0091417 | 12/2018 | McCaffrey et al.   | N/A | N/A |
| 2019/0111202 | 12/2018 | Falkovich          | N/A | N/A |
| 2019/0117896 | 12/2018 | Booth et al.       | N/A | N/A |
| 2019/0117897 | 12/2018 | Avery et al.       | N/A | N/A |
| 2019/0125226 | 12/2018 | Koya et al.        | N/A | N/A |
| 2019/0175828 | 12/2018 | List et al.        | N/A | N/A |
| 2019/0262535 | 12/2018 | Shubinsky et al.   | N/A | N/A |
| 2019/0275243 | 12/2018 | Deck et al.        | N/A | N/A |
| 2019/0275249 | 12/2018 | von Campenhausen   | N/A | N/A |
| 2019/0282751 | 12/2018 | Della Bidia        | N/A | N/A |
| 2019/0290845 | 12/2018 | List               | N/A | N/A |
| 2019/0298485 | 12/2018 | Forsell            | N/A | N/A |
| 2019/0298912 | 12/2018 | Spencer et al.     | N/A | N/A |
| 2019/0298914 | 12/2018 | Kamen et al.       | N/A | N/A |
| 2019/0298916 | 12/2018 | List               | N/A | N/A |
| 2019/0298918 | 12/2018 | Jallon             | N/A | N/A |
| 2019/0298921 | 12/2018 | Stafford           | N/A | N/A |
| 2019/0298925 | 12/2018 | Cowe et al.        | N/A | N/A |
| 2019/0307943 | 12/2018 | Franano et al.     | N/A | N/A |
| 2019/0307952 | 12/2018 | Butler et al.      | N/A | N/A |
| 2019/0307954 | 12/2018 | Klemm et al.       | N/A | N/A |
| 2019/0307955 | 12/2018 | Levesque et al.    | N/A | N/A |
| 2019/0307970 | 12/2018 | Karen et al.       | N/A | N/A |
| 2019/0314572 | 12/2018 | Yang               | N/A | N/A |
| 2019/0321260 | 12/2018 | Grant et al.       | N/A | N/A |
| 2019/0321544 | 12/2018 | List               | N/A | N/A |
| 2019/0321545 | 12/2018 | Saint              | N/A | N/A |
| 2019/0321546 | 12/2018 | Michaud et al.     | N/A | N/A |
| 2019/0321548 | 12/2018 | Cowan              | N/A | N/A |
| 2019/0321552 | 12/2018 | DiPema et al.      | N/A | N/A |
| 2019/0328963 | 12/2018 | Wolff et al.       | N/A | N/A |
| 2019/0336078 | 12/2018 | Dang et al.        | N/A | N/A |
| 2019/0336678 | 12/2018 | Rule               | N/A | N/A |
| 2019/0336679 | 12/2018 | Staub et al.       | N/A | N/A |
| 2019/0336681 | 12/2018 | Kamen et al.       | N/A | N/A |
| 2019/0336683 | 12/2018 | O'Connor et al.    | N/A | N/A |
| 2019/0341149 | 12/2018 | Chiu et al.        | N/A | N/A |
| 2019/0343434 | 12/2018 | Varsavsky et al.   | N/A | N/A |
| 2019/0344009 | 12/2018 | Damiano et al.     | N/A | N/A |
| 2019/0344010 | 12/2018 | Pizzochero et al.  | N/A | N/A |
| 2019/0344057 | 12/2018 | Cima et al.        | N/A | N/A |
| 2019/0350501 | 12/2018 | Blomquist et al.   | N/A | N/A |
| 2019/0351131 | 12/2018 | Butterfield et al. | N/A | N/A |
| 2019/0351132 | 12/2018 | Pippin et al.      | N/A | N/A |
| 2019/0351133 | 12/2018 | Grant, Jr. et al.  | N/A | N/A |
| 2019/0351134 | 12/2018 | Cook et al.        | N/A | N/A |
| 2019/0351135 | 12/2018 | Naftalovitz et al. | N/A | N/A |

|              |         |                    |     |     |
|--------------|---------|--------------------|-----|-----|
| 2019/0351138 | 12/2018 | Bhandar et al.     | N/A | N/A |
| 2019/0351143 | 12/2018 | Egloff et al.      | N/A | N/A |
| 2019/0351209 | 12/2018 | Butziger et al.    | N/A | N/A |
| 2019/0358393 | 12/2018 | Marbet             | N/A | N/A |
| 2019/0358395 | 12/2018 | Olson et al.       | N/A | N/A |
| 2019/0358437 | 12/2018 | Schwartz et al.    | N/A | N/A |
| 2019/0365282 | 12/2018 | Gibson             | N/A | N/A |
| 2019/0365985 | 12/2018 | Zidon et al.       | N/A | N/A |
| 2019/0365986 | 12/2018 | Coiner et al.      | N/A | N/A |
| 2019/0365987 | 12/2018 | Gibson et al.      | N/A | N/A |
| 2019/0365993 | 12/2018 | Staub et al.       | N/A | N/A |
| 2019/0366002 | 12/2018 | Verlaak et al.     | N/A | N/A |
| 2019/0366011 | 12/2018 | Ring               | N/A | N/A |
| 2019/0366012 | 12/2018 | Gross et al.       | N/A | N/A |
| 2019/0368484 | 12/2018 | Chappel et al.     | N/A | N/A |
| 2019/0374434 | 12/2018 | Kamdar et al.      | N/A | N/A |
| 2019/0374706 | 12/2018 | Cabiri et al.      | N/A | N/A |
| 2019/0374707 | 12/2018 | Damestani et al.   | N/A | N/A |
| 2019/0374708 | 12/2018 | Cardinali et al.   | N/A | N/A |
| 2019/0374709 | 12/2018 | Cole et al.        | N/A | N/A |
| 2019/0374711 | 12/2018 | Deliwala           | N/A | N/A |
| 2019/0374714 | 12/2018 | Rioux et al.       | N/A | N/A |
| 2019/0374719 | 12/2018 | Cabiri et al.      | N/A | N/A |
| 2019/0374757 | 12/2018 | Verhoeven et al.   | N/A | N/A |
| 2019/0381238 | 12/2018 | Stonecipher et al. | N/A | N/A |
| 2019/0381239 | 12/2018 | Cabiri et al.      | N/A | N/A |
| 2019/0381241 | 12/2018 | Bryant et al.      | N/A | N/A |
| 2019/0388609 | 12/2018 | Lanigan et al.     | N/A | N/A |
| 2019/0388612 | 12/2018 | Schramm            | N/A | N/A |
| 2019/0388614 | 12/2018 | Gyrn et al.        | N/A | N/A |
| 2019/0388615 | 12/2018 | Sonderegger et al. | N/A | N/A |
| 2019/0392938 | 12/2018 | Mermet             | N/A | N/A |
| 2020/0001004 | 12/2019 | Kondo              | N/A | N/A |
| 2020/0001005 | 12/2019 | Politis et al.     | N/A | N/A |
| 2020/0001006 | 12/2019 | Pizzochero et al.  | N/A | N/A |
| 2020/0001007 | 12/2019 | Miesel et al.      | N/A | N/A |
| 2020/0009315 | 12/2019 | Brouet et al.      | N/A | N/A |
| 2020/0009317 | 12/2019 | Cronenberg et al.  | N/A | N/A |
| 2020/0009318 | 12/2019 | Kamen et al.       | N/A | N/A |
| 2020/0009319 | 12/2019 | Ludolph            | N/A | N/A |
| 2020/0009324 | 12/2019 | Barrows et al.     | N/A | N/A |
| 2020/0009331 | 12/2019 | Kamen et al.       | N/A | N/A |
| 2020/0016328 | 12/2019 | Cane' et al.       | N/A | N/A |
| 2020/0016329 | 12/2019 | Schabbach et al.   | N/A | N/A |
| 2020/0016330 | 12/2019 | Kapas et al.       | N/A | N/A |
| 2020/0016333 | 12/2019 | Soares et al.      | N/A | N/A |
| 2020/0016335 | 12/2019 | DiPerna et al.     | N/A | N/A |
| 2020/0016336 | 12/2019 | Patek et al.       | N/A | N/A |
| 2020/0023119 | 12/2019 | Barnes et al.      | N/A | N/A |
| 2020/0023121 | 12/2019 | Thomas et al.      | N/A | N/A |

|              |         |                    |     |     |
|--------------|---------|--------------------|-----|-----|
| 2020/0023122 | 12/2019 | McCullough et al.  | N/A | N/A |
| 2020/0023123 | 12/2019 | O'Connor et al.    | N/A | N/A |
| 2020/0023129 | 12/2019 | Day et al.         | N/A | N/A |
| 2020/0025184 | 12/2019 | Gyory              | N/A | N/A |
| 2020/0027541 | 12/2019 | Xavier et al.      | N/A | N/A |
| 2020/0028914 | 12/2019 | Xavier et al.      | N/A | N/A |
| 2020/0030528 | 12/2019 | Burke et al.       | N/A | N/A |
| 2020/0030529 | 12/2019 | DiPerna et al.     | N/A | N/A |
| 2020/0030530 | 12/2019 | Huang et al.       | N/A | N/A |
| 2020/0030531 | 12/2019 | Day et al.         | N/A | N/A |
| 2020/0030532 | 12/2019 | Day et al.         | N/A | N/A |
| 2020/0030533 | 12/2019 | Day et al.         | N/A | N/A |
| 2020/0030590 | 12/2019 | Buchman et al.     | N/A | N/A |
| 2020/0030592 | 12/2019 | Cheche             | N/A | N/A |
| 2020/0035355 | 12/2019 | Xavier et al.      | N/A | N/A |
| 2020/0038588 | 12/2019 | Varsavsky et al.   | N/A | N/A |
| 2020/0043588 | 12/2019 | Mougiakakou et al. | N/A | N/A |
| 2020/0046904 | 12/2019 | Schader et al.     | N/A | N/A |
| 2020/0054822 | 12/2019 | Dewey              | N/A | N/A |
| 2020/0054825 | 12/2019 | Kamen et al.       | N/A | N/A |
| 2020/0054826 | 12/2019 | Diianni et al.     | N/A | N/A |
| 2020/0054832 | 12/2019 | Jeong et al.       | N/A | N/A |
| 2020/0061285 | 12/2019 | Reeves             | N/A | N/A |
| 2020/0061287 | 12/2019 | Chappel et al.     | N/A | N/A |
| 2020/0069865 | 12/2019 | Day et al.         | N/A | N/A |
| 2020/0069869 | 12/2019 | Grant et al.       | N/A | N/A |
| 2020/0069871 | 12/2019 | Yavorsky et al.    | N/A | N/A |
| 2020/0069873 | 12/2019 | Pizzochero et al.  | N/A | N/A |
| 2020/0069875 | 12/2019 | Nazzaro et al.     | N/A | N/A |
| 2020/0077340 | 12/2019 | Kruse              | N/A | N/A |
| 2020/0077948 | 12/2019 | Schmid             | N/A | N/A |
| 2020/0078511 | 12/2019 | Focht et al.       | N/A | N/A |
| 2020/0078513 | 12/2019 | Wei                | N/A | N/A |
| 2020/0086041 | 12/2019 | Fuchs et al.       | N/A | N/A |
| 2020/0086042 | 12/2019 | Kamen et al.       | N/A | N/A |
| 2020/0086043 | 12/2019 | Saint              | N/A | N/A |
| 2020/0086044 | 12/2019 | Streit et al.      | N/A | N/A |
| 2020/0086045 | 12/2019 | Azapagic et al.    | N/A | N/A |
| 2020/0086051 | 12/2019 | Grygus et al.      | N/A | N/A |
| 2020/0093980 | 12/2019 | McDermott et al.   | N/A | N/A |
| 2020/0093984 | 12/2019 | Shor et al.        | N/A | N/A |
| 2020/0098463 | 12/2019 | Arunachalam et al. | N/A | N/A |
| 2020/0098464 | 12/2019 | Velado et al.      | N/A | N/A |
| 2020/0101218 | 12/2019 | Shapley et al.     | N/A | N/A |
| 2020/0101219 | 12/2019 | Wang et al.        | N/A | N/A |
| 2020/0101222 | 12/2019 | Lintereur et al.   | N/A | N/A |
| 2020/0101223 | 12/2019 | Lintereur et al.   | N/A | N/A |
| 2020/0101224 | 12/2019 | Lintereur et al.   | N/A | N/A |
| 2020/0101225 | 12/2019 | O'Connor et al.    | N/A | N/A |
| 2020/0101226 | 12/2019 | Rosinko et al.     | N/A | N/A |

|              |         |                    |     |     |
|--------------|---------|--------------------|-----|-----|
| 2020/0108201 | 12/2019 | Ben-David et al.   | N/A | N/A |
| 2020/0108204 | 12/2019 | Mazlish et al.     | N/A | N/A |
| 2020/0111556 | 12/2019 | Schmidlin et al.   | N/A | N/A |
| 2020/0113515 | 12/2019 | O'Connor et al.    | N/A | N/A |
| 2020/0114064 | 12/2019 | Reeves             | N/A | N/A |
| 2020/0114068 | 12/2019 | Schmidlin et al.   | N/A | N/A |
| 2020/0114069 | 12/2019 | Searle et al.      | N/A | N/A |
| 2020/0114072 | 12/2019 | Addiego et al.     | N/A | N/A |
| 2020/0114075 | 12/2019 | Morrow et al.      | N/A | N/A |
| 2020/0114076 | 12/2019 | Ulrich et al.      | N/A | N/A |
| 2020/0114080 | 12/2019 | Barmaimon et al.   | N/A | N/A |
| 2020/0118676 | 12/2019 | Spohn et al.       | N/A | N/A |
| 2020/0121848 | 12/2019 | Schmidlin et al.   | N/A | N/A |
| 2020/0121849 | 12/2019 | Christenson et al. | N/A | N/A |
| 2020/0121850 | 12/2019 | Christenson et al. | N/A | N/A |
| 2020/0121854 | 12/2019 | Norton et al.      | N/A | N/A |
| 2020/0121937 | 12/2019 | Yoder et al.       | N/A | N/A |
| 2020/0129692 | 12/2019 | Kim et al.         | N/A | N/A |
| 2020/0135323 | 12/2019 | Bazargan           | N/A | N/A |
| 2020/0138852 | 12/2019 | Chattaraj et al.   | N/A | N/A |
| 2020/0138911 | 12/2019 | Joseph et al.      | N/A | N/A |
| 2020/0139137 | 12/2019 | Crawford           | N/A | N/A |
| 2020/0146938 | 12/2019 | Bourelle et al.    | N/A | N/A |
| 2020/0147298 | 12/2019 | Traverso et al.    | N/A | N/A |
| 2020/0147303 | 12/2019 | Lee                | N/A | N/A |
| 2020/0147304 | 12/2019 | Crouther et al.    | N/A | N/A |
| 2020/0147305 | 12/2019 | Estes              | N/A | N/A |
| 2020/0147309 | 12/2019 | Quinn et al.       | N/A | N/A |
| 2020/0155755 | 12/2019 | Chaves et al.      | N/A | N/A |
| 2020/0155757 | 12/2019 | Gregory et al.     | N/A | N/A |
| 2020/0155758 | 12/2019 | Reeves             | N/A | N/A |
| 2020/0164142 | 12/2019 | Poetschke          | N/A | N/A |
| 2020/0164143 | 12/2019 | Cardinali et al.   | N/A | N/A |
| 2020/0164159 | 12/2019 | Chattaraj et al.   | N/A | N/A |
| 2020/0164199 | 12/2019 | Gerlach et al.     | N/A | N/A |
| 2020/0168316 | 12/2019 | Kamen              | N/A | N/A |
| 2020/0171236 | 12/2019 | McCullough et al.  | N/A | N/A |
| 2020/0171294 | 12/2019 | Turner et al.      | N/A | N/A |
| 2020/0179592 | 12/2019 | Adams et al.       | N/A | N/A |
| 2020/0179594 | 12/2019 | Yodfat et al.      | N/A | N/A |
| 2020/0179595 | 12/2019 | McDermott et al.   | N/A | N/A |
| 2020/0179596 | 12/2019 | Dechelette et al.  | N/A | N/A |
| 2020/0179598 | 12/2019 | Penake et al.      | N/A | N/A |
| 2020/0179602 | 12/2019 | Mazlish            | N/A | N/A |
| 2020/0179603 | 12/2019 | Rosinko            | N/A | N/A |
| 2020/0179604 | 12/2019 | Friedli            | N/A | N/A |
| 2020/0179610 | 12/2019 | Bar-El et al.      | N/A | N/A |
| 2020/0188578 | 12/2019 | Bar-El et al.      | N/A | N/A |
| 2020/0188580 | 12/2019 | Gregory et al.     | N/A | N/A |
| 2020/0188581 | 12/2019 | Diianni et al.     | N/A | N/A |

|              |         |                   |     |     |
|--------------|---------|-------------------|-----|-----|
| 2020/0188585 | 12/2019 | Petisce et al.    | N/A | N/A |
| 2020/0188587 | 12/2019 | Sluggett et al.   | N/A | N/A |
| 2020/0188588 | 12/2019 | Estes             | N/A | N/A |
| 2020/0188608 | 12/2019 | Yigal et al.      | N/A | N/A |
| 2020/0197600 | 12/2019 | Chow et al.       | N/A | N/A |
| 2020/0197603 | 12/2019 | Cowe et al.       | N/A | N/A |
| 2020/0197604 | 12/2019 | Fried             | N/A | N/A |
| 2020/0197621 | 12/2019 | Quinn et al.      | N/A | N/A |
| 2020/0197628 | 12/2019 | McCullough et al. | N/A | N/A |
| 2020/0206417 | 12/2019 | Yodfat et al.     | N/A | N/A |
| 2020/0206418 | 12/2019 | Gonnelli et al.   | N/A | N/A |
| 2020/0206422 | 12/2019 | Cassim            | N/A | N/A |
| 2020/0206429 | 12/2019 | Hering et al.     | N/A | N/A |
| 2020/0214625 | 12/2019 | Hooven et al.     | N/A | N/A |
| 2020/0215264 | 12/2019 | Searle et al.     | N/A | N/A |
| 2020/0215273 | 12/2019 | Gibson et al.     | N/A | N/A |
| 2020/0222624 | 12/2019 | Destefano et al.  | N/A | N/A |
| 2020/0222625 | 12/2019 | Cabiri et al.     | N/A | N/A |
| 2020/0230313 | 12/2019 | Mojarrad et al.   | N/A | N/A |
| 2020/0230314 | 12/2019 | Kondo et al.      | N/A | N/A |
| 2020/0238003 | 12/2019 | Yigal et al.      | N/A | N/A |
| 2020/0238004 | 12/2019 | McCullough        | N/A | N/A |
| 2020/0238006 | 12/2019 | Groszmann et al.  | N/A | N/A |
| 2020/0238012 | 12/2019 | Bar-El et al.     | N/A | N/A |
| 2020/0246538 | 12/2019 | Bar-El et al.     | N/A | N/A |
| 2020/0246541 | 12/2019 | Neftel et al.     | N/A | N/A |
| 2020/0253632 | 12/2019 | Chong et al.      | N/A | N/A |
| 2020/0254172 | 12/2019 | Forster et al.    | N/A | N/A |
| 2020/0254174 | 12/2019 | Kruse et al.      | N/A | N/A |
| 2020/0254176 | 12/2019 | Rytz et al.       | N/A | N/A |
| 2020/0261002 | 12/2019 | Pace              | N/A | N/A |
| 2020/0261642 | 12/2019 | Ben-David et al.  | N/A | N/A |
| 2020/0261643 | 12/2019 | Boyaval et al.    | N/A | N/A |
| 2020/0261645 | 12/2019 | Kamen et al.      | N/A | N/A |
| 2020/0261658 | 12/2019 | Farris et al.     | N/A | N/A |
| 2020/0268962 | 12/2019 | Gamelin           | N/A | N/A |
| 2020/0268975 | 12/2019 | Kim et al.        | N/A | N/A |
| 2020/0272310 | 12/2019 | Vik et al.        | N/A | N/A |
| 2020/0276384 | 12/2019 | Cabiri et al.     | N/A | N/A |
| 2020/0276386 | 12/2019 | Karen et al.      | N/A | N/A |
| 2020/0282131 | 12/2019 | Nazzaro           | N/A | N/A |
| 2020/0289743 | 12/2019 | Chiu et al.       | N/A | N/A |
| 2020/0289745 | 12/2019 | Harris et al.     | N/A | N/A |
| 2020/0297920 | 12/2019 | McLaughlin        | N/A | N/A |
| 2020/0297923 | 12/2019 | Montalvo et al.   | N/A | N/A |
| 2020/0297927 | 12/2019 | Conrath et al.    | N/A | N/A |
| 2020/0306444 | 12/2019 | Politis et al.    | N/A | N/A |
| 2020/0306446 | 12/2019 | Kamen et al.      | N/A | N/A |
| 2020/0306448 | 12/2019 | Schmid            | N/A | N/A |
| 2020/0316290 | 12/2019 | Bourelle et al.   | N/A | N/A |

|              |         |                   |     |     |
|--------------|---------|-------------------|-----|-----|
| 2020/0316291 | 12/2019 | Gibson et al.     | N/A | N/A |
| 2020/0316314 | 12/2019 | Buri et al.       | N/A | N/A |
| 2020/0321094 | 12/2019 | Saint et al.      | N/A | N/A |
| 2020/0324048 | 12/2019 | O'Connor et al.   | N/A | N/A |
| 2020/0324101 | 12/2019 | Hartmann et al.   | N/A | N/A |
| 2020/0330679 | 12/2019 | Cronenberg et al. | N/A | N/A |
| 2020/0330680 | 12/2019 | Deck              | N/A | N/A |
| 2020/0330701 | 12/2019 | Cole et al.       | N/A | N/A |
| 2020/0335194 | 12/2019 | Jacobson et al.   | N/A | N/A |
| 2020/0338257 | 12/2019 | Hooven et al.     | N/A | N/A |
| 2020/0338262 | 12/2019 | Kamen et al.      | N/A | N/A |
| 2020/0338264 | 12/2019 | Allis et al.      | N/A | N/A |
| 2020/0338266 | 12/2019 | Estes             | N/A | N/A |
| 2021/0060241 | 12/2020 | Kamen et al.      | N/A | N/A |
| 2021/0178074 | 12/2020 | Anderson et al.   | N/A | N/A |
| 2021/0180583 | 12/2020 | Gray              | N/A | N/A |
| 2021/0180584 | 12/2020 | Kamen et al.      | N/A | N/A |
| 2021/0190063 | 12/2020 | Gray              | N/A | N/A |
| 2021/0268174 | 12/2020 | Estes et al.      | N/A | N/A |
| 2021/0270255 | 12/2020 | Gray              | N/A | N/A |
| 2021/0293232 | 12/2020 | Kamen et al.      | N/A | N/A |
| 2021/0321914 | 12/2020 | Brister et al.    | N/A | N/A |
| 2021/0330878 | 12/2020 | Margairaz et al.  | N/A | N/A |
| 2021/0330885 | 12/2020 | Halac et al.      | N/A | N/A |
| 2021/0396221 | 12/2020 | Kamen et al.      | N/A | N/A |
| 2022/0001106 | 12/2021 | DiPerna et al.    | N/A | N/A |
| 2022/0023544 | 12/2021 | Anderson et al.   | N/A | N/A |
| 2022/0105261 | 12/2021 | Yodfat et al.     | N/A | N/A |
| 2022/0184303 | 12/2021 | DiPerna et al.    | N/A | N/A |
| 2022/0249769 | 12/2021 | Kamen et al.      | N/A | N/A |
| 2022/0275796 | 12/2021 | Kamen et al.      | N/A | N/A |
| 2022/0280715 | 12/2021 | Gray et al.       | N/A | N/A |
| 2023/0226271 | 12/2022 | Lanigan et al.    | N/A | N/A |
| 2023/0226273 | 12/2022 | Kamen et al.      | N/A | N/A |
| 2023/0226274 | 12/2022 | Kamen et al.      | N/A | N/A |
| 2023/0372606 | 12/2022 | Kamen et al.      | N/A | N/A |
| 2024/0033429 | 12/2023 | Jennewine         | N/A | N/A |
| 2024/0066214 | 12/2023 | Kamen et al.      | N/A | N/A |

## FOREIGN PATENT DOCUMENTS

| Patent No. | Application Date | Country | CPC |
|------------|------------------|---------|-----|
| 3257533    | 12/2016          | EP      | N/A |
| 3305349    | 12/2017          | EP      | N/A |
| 2995330    | 12/2020          | EP      | N/A |
| 3650370    | 12/2020          | EP      | N/A |
| 3760253    | 12/2021          | EP      | N/A |
| 3998095    | 12/2021          | EP      | N/A |
| 2611478    | 12/2021          | EP      | N/A |
| 3470100    | 12/2021          | EP      | N/A |

|            |         |    |     |
|------------|---------|----|-----|
| 2575935    | 12/2022 | EP | N/A |
| 3998095    | 12/2023 | EP | N/A |
| 3656699    | 12/2023 | EP | N/A |
| 4385551    | 12/2023 | EP | N/A |
| 2009010399 | 12/2008 | WO | N/A |
| 2010051079 | 12/2009 | WO | N/A |
| 2010084268 | 12/2009 | WO | N/A |
| 2012123274 | 12/2011 | WO | N/A |
| 2015094945 | 12/2014 | WO | N/A |
| 2018129519 | 12/2017 | WO | N/A |
| 2018210972 | 12/2017 | WO | N/A |
| 2018215465 | 12/2017 | WO | N/A |
| 2018218082 | 12/2017 | WO | N/A |
| 2018222521 | 12/2017 | WO | N/A |
| 2018232171 | 12/2017 | WO | N/A |
| 2019018838 | 12/2018 | WO | N/A |
| 2019022950 | 12/2018 | WO | N/A |
| 2019022951 | 12/2018 | WO | N/A |
| 2019038751 | 12/2018 | WO | N/A |
| 2019043702 | 12/2018 | WO | N/A |
| 2019067386 | 12/2018 | WO | N/A |
| 2019070472 | 12/2018 | WO | N/A |
| 2019074579 | 12/2018 | WO | N/A |
| 2019079868 | 12/2018 | WO | N/A |
| 2019081947 | 12/2018 | WO | N/A |
| 2019186375 | 12/2018 | WO | N/A |
| 2019191222 | 12/2018 | WO | N/A |
| 2019193089 | 12/2018 | WO | N/A |
| 2019197360 | 12/2018 | WO | N/A |
| 2019197361 | 12/2018 | WO | N/A |
| 2019200198 | 12/2018 | WO | N/A |
| 2019213218 | 12/2018 | WO | N/A |
| 2019228895 | 12/2018 | WO | N/A |
| 2019229686 | 12/2018 | WO | N/A |
| 2020005107 | 12/2019 | WO | N/A |
| 2020008017 | 12/2019 | WO | N/A |
| 2020011572 | 12/2019 | WO | N/A |
| 2020012132 | 12/2019 | WO | N/A |
| 2020012308 | 12/2019 | WO | N/A |
| 2020013691 | 12/2019 | WO | N/A |
| 2020016172 | 12/2019 | WO | N/A |
| 2020025484 | 12/2019 | WO | N/A |
| 2020028009 | 12/2019 | WO | N/A |
| 2020043459 | 12/2019 | WO | N/A |
| 2020046889 | 12/2019 | WO | N/A |
| 2020052723 | 12/2019 | WO | N/A |
| 2020055785 | 12/2019 | WO | N/A |
| 2020068623 | 12/2019 | WO | N/A |
| 2020069926 | 12/2019 | WO | N/A |
| 2020072233 | 12/2019 | WO | N/A |

|            |         |    |     |
|------------|---------|----|-----|
| 2020072234 | 12/2019 | WO | N/A |
| 2020072235 | 12/2019 | WO | N/A |
| 2020074988 | 12/2019 | WO | N/A |
| 2020075042 | 12/2019 | WO | N/A |
| 2020104872 | 12/2019 | WO | N/A |
| 2020109409 | 12/2019 | WO | N/A |
| 2020109417 | 12/2019 | WO | N/A |
| 2020112515 | 12/2019 | WO | N/A |
| 2020118165 | 12/2019 | WO | N/A |
| 2020120511 | 12/2019 | WO | N/A |
| 2020127181 | 12/2019 | WO | N/A |
| 2020141412 | 12/2019 | WO | N/A |
| 2020144270 | 12/2019 | WO | N/A |
| 2020146306 | 12/2019 | WO | N/A |
| 2020148581 | 12/2019 | WO | N/A |

## OTHER PUBLICATIONS

International Application No. PCT/EP2012/053722 International Preliminary Report on Patentability issued Sep. 17, 2013. cited by applicant

International Application No. PCT/EP2012/053722 International Search Report completed Mar. 29, 2012. cited by applicant

International Application No. PCT/EP2012/053722 Written Opinion completed Mar. 29, 2012. cited by applicant

U.S. Appl. No. 14/003,193 Office Action dated Apr. 22, 2016. cited by applicant

U.S. Appl. No. 14/003,193 Office Action dated Jun. 1, 2017. cited by applicant

U.S. Appl. No. 14/003,193 Office Action dated May 18, 2018. cited by applicant

U.S. Appl. No. 14/003,193 Office Action dated Oct. 5, 2017. cited by applicant

U.S. Appl. No. 14/003,193 Office Action dated Sep. 13, 2016. cited by applicant

U.S. Appl. No. 14/003,193 Notice of Allowance dated Sep. 25, 2018. cited by applicant

*Primary Examiner:* McEvoy; Thomas

*Attorney, Agent or Firm:* TAFT STETTINIUS HOLLISTER LLP

## Background/Summary

CROSS-REFERENCE (1) This application is a continuation of U.S. patent application Ser. No. 16/265,832, filed Feb. 1, 2019, which is a continuation application of U.S. patent application Ser. No. 14/003,193, filed Nov. 12, 2013, which is a U.S. National Phase of International Application No. PCT/EP2012/053722, filed Mar. 5, 2012, which claims the benefit of U.S. Provisional Application No. 61/452,836, filed Mar. 15, 2011 and EP application Ser. No. 11/158,118.7, filed Mar. 14, 2011, each of which applications is incorporated herein by reference in its entirety and to which applications we claim priority.

## BACKGROUND OF THE INVENTION

(1) The present invention relates to an inserter system, in particular a single use inserter system for placing a transcutaneous device on or near the skin of a patient.

(2) Inserter devices that require manual loading of a drive prior to insertion through the skin of a patient typically have a carrier part carrying the transcutaneous device, wherein the carrier part is in



an initial position prior to use, i.e. during storage and transportation, wherein the drive is unbiased/unloaded or substantially unbiased. During transportation, the inserter device may be subject to shock or bumps, which may lead to movement of the unbiased carrier part possibly causing breakage of the packaging, e.g. separation of a seal or cover sheet from the housing. Further, shocks or bumps affecting the inserter system may result in displacement of the transcutaneous device in relation to the carrier part. Displacement of the transcutaneous device may lead to discomfort during insertion or even malfunction of the inserter system.

(3) Typically, preloading of inserter devices with spring elements made of a plastic material, such as POM, is not suitable due to the material properties of the spring element material. Accordingly for inserter devices with plastic spring elements, the drive unit is typically unloaded or substantially unloaded in the initial or transport position, thus requiring loading of a drive unit of the inserter device prior to placing or inserting a transcutaneous part of the inserter device.

#### SUMMARY

(4) There is a need for inserter systems with capability of withstanding shocks and bumps, e.g. during transportation or relocation.

(5) Accordingly, an inserter system is provided, the inserter system being a single use inserter system comprising a housing part and a lid part, the inserter system comprising a carrier part in an initial position in the housing and a drive unit comprising a first drive part attached to the housing, the drive unit comprising at least one spring element supported by the first drive part, wherein the drive unit is configured for moving the carrier part from a first position to a second position in relation to the housing in an insertion direction along a first axis. The inserter system may comprise a transcutaneous device having a proximal surface and a distal surface, wherein the transcutaneous device is detachably attached to the carrier part, and the inserter system may comprise at least one transport protection element preventing movement of the carrier part in a direction along the first axis thereby supporting the carrier part in the initial position.

(6) The inserter system provides an improved inserter system with increased strength and which is able to withstand shocks and bumps typically experienced during transport and storage without damaging the inserter system.

---

## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other features and advantages of the present invention will become readily apparent to those skilled in the art by the following detailed description of exemplary embodiments thereof with reference to the attached drawings, in which:

(2) FIG. 1 is an exploded view of an exemplary inserter system,

(3) FIG. 2 is a different exploded view of the inserter system in FIG. 1,

(4) FIG. 3 shows a cross section of the inserter system in FIG. 1,

(5) FIG. 4 shows the inserter system of FIG. 1 perpendicular to the insertion direction,

(6) FIG. 5 shows a perspective view of a first unit of an inserter system,

(7) FIG. 6 shows another perspective view of the first unit in FIG. 5,

(8) FIG. 7 shows a cross section of an inserter system,

(9) FIG. 8 shows the inserter system of FIG. 7,

(10) FIG. 9 shows a perspective view of the inserter system in FIG. 7, and

(11) FIG. 10 shows a locking member.

(12) The figures are schematic and simplified for clarity, and they merely show details which are essential to the understanding of the invention, while other details have been left out. Throughout, the same reference numerals are used for identical or corresponding parts.

### DETAILED DESCRIPTION OF THE INVENTION

(13) The inserter system of the present invention alleviates the problems experienced with inserter devices with unbiased drive units.

(14) The inserter system comprises a housing part and optionally a lid part. The housing part has a sidewall with a first end and a second end. A first opening and/or a second opening may be provided at the first end and second end, respectively. When assembled, the lid part may cover the first opening at the first end.

(15) The inserter system may comprise a gas permeable seal or cover sheet. The seal may cover the second opening at the second end of the housing part. The housing part, the seal, and optionally the lid part may define a closed cavity accommodating the transcutaneous device in a sterile environment prior to use. The gas permeable seal allows sterilization of the inserter system upon assembly.

(16) The inserter system comprises a transcutaneous device comprising a base with a distal surface and a proximal surface, the transcutaneous device comprising a transcutaneous element, such as a cannula and/or a sensor element, mounted to the base extending from the proximal surface facing the skin of a patient. The proximal surface may be an adhesive surface for securing the transcutaneous device to a base plate and/or on the skin of the patient. A mounting pad with an adhesive surface may be attached to the proximal surface of the base of the transcutaneous device. The inserter system may comprise a protective layer covering the adhesive surface. The transcutaneous device may comprise locking elements for locking engagement with a base plate.

(17) Prior to use, the carrier part is in an initial position. Prior to inserting the transcutaneous device, a user has to load the inserter system. During loading of the inserter system, the carrier part and the transcutaneous device are moved by the user from the initial position to a first position also referred to as a loaded position or retracted position, e.g. in a retraction direction opposite the insertion direction. By moving the carrier part, e.g. along the first axis and/or around the first axis, to the first position, the drive unit connected to or supporting on the carrier part is biased or loaded for moving the carrier part and the transcutaneous device from the first position to a second position also referred to as an injection position. In the first position, the carrier part is releasably locked by one or more first locking elements, such as a first primary locking element and/or a first secondary locking element on the carrier part. In the first position, the first locking elements may support on a stationary part of the inserter system including the housing part and/or the first drive part. After loading, the inserter system is positioned on the skin and the carrier part is released. Upon release of the first locking element(s), the loaded drive unit moves the carrier part and the transcutaneous device in the insertion direction to the second position placing the transcutaneous element under the skin. After insertion, the carrier part and housing part are removed and the transcutaneous element, optionally including tubing, is ready for use.

(18) The carrier part of the inserter system may comprise a needle hub with a hub base and optionally an insertion needle secured to the hub base, wherein the transcutaneous device is detachably attached to the needle hub. The carrier part may comprise a carrier base, wherein the needle hub is releasably attached to the carrier base. The inserter device may be configured for releasing the needle hub from the carrier base, e.g. in the second position. The inserter device may be configured for moving the needle hub to a third position in the extraction direction upon release from the carrier base in the second position.

(19) The carrier part, with or without an insertion needle, and the drive unit may be molded in a single first unit reducing assembly costs.

(20) The inserter system comprises a stationary part. The stationary part is the parts of the inserter system that are stationary in relation to the skin of a patient when the transcutaneous device is inserted. The stationary part comprises the housing part and at least the first drive part(s).

(21) The at least one transport protection element provides a releasable support or locking of the carrier part and/or transcutaneous device in the initial position, thereby preventing or limiting movement of the carrier part and/or transcutaneous device in the insertion direction and/or in the

extraction direction opposite the insertion direction along the first axis. Thus, the transport protection elements may provide that the transcutaneous device in the initial position is substantially fixed or secured in relation to the carrier part thereby avoiding user discomfort or malfunction of the inserter system during use. Further, the transport protection element(s) may provide fixation of the carrier part and/or the transcutaneous device in relation to the stationary part reducing or eliminating the risk of breaking the seal or cover sheet during transportation.

(22) The at least one transport protection element may comprise at least one bridge between the carrier part and the stationary part of the inserter system. The at least one bridge has dimensions or weakened sections that enable a user to break the bridges when loading the inserter system in order to move the carrier part to the first position in relation to the stationary part. At the same time, the bridge dimensions are selected to withstand forces resulting from transport bumps or shocks, thereby securing the carrier part in the initial position.

(23) A bridge may comprise one or more weakened sections. A weakened section may have a minimum width in the range from about 0.1 mm to about 2.0 mm, such as in the range from 0.3 mm to 1.5 mm, e.g. about 1.0 mm. A weakened section may have a minimum thickness in the range from about 0.1 mm to about 2.0 mm, such as in the range from 0.3 mm to 1.5 mm, e.g. about 0.5 mm. A weakened section of a bridge may have a minimum cross sectional area in the range from 0.1 mm.<sup>2</sup> to about 4.0 mm.<sup>2</sup>, such as in the range from about 0.3 mm.<sup>2</sup> to about 2.0 mm.<sup>2</sup>, e.g. 0.5 mm.<sup>2</sup>, 0.8 mm.<sup>2</sup> or 1.0 mm.<sup>2</sup>

(24) The first unit may be made of a suitable polymer material such as polyoxymethylene (POM).

(25) The at least one transport protection element may comprise a locking member detachably mounted in the housing. The locking member may engage with or supporting on the carrier part and/or a stationary part of the inserter system, such that movement of the carrier part in a direction, e.g. the insertion direction and/or the extraction direction, along the first axis in relation to the stationary part is limited or substantially prevented, thereby supporting the carrier part in the initial position. The locking member may limit or substantially prevent movement of the transcutaneous device in the extraction direction and/or in the insertion direction.

(26) The at least one transport protection element may comprise at least one support element formed in the lid part. The at least one support element may include a first support element for supporting the carrier part and the transcutaneous device in the initial position. The first support element may have a first end positioned adjacent to the proximal surface of the transcutaneous device limiting or substantially preventing movement of the transcutaneous device in the insertion direction.

(27) A first support element may be formed as a cylindrical tube having a first end and a suitable cross section, such as circular with a first diameter  $d_{sub.1}$ , rectangular, oval, or any other suitable shape. The first support element may extend perpendicular to the first axis. The first support element may be arranged to support the transcutaneous device in the initial position by the first end of the first support element forming a stop member in the insertion direction. The distance between the first end of the first support element and the proximal surface of the transcutaneous device should be small enough to prevent displacement of the transcutaneous device in relation to the carrier part and/or the stationary part in the insertion direction prior to use, e.g. during storage or transportation. The distance between the first end of the first support element and the proximal surface of the transcutaneous device may be less than 2 mm, such as less than 1 mm.

(28) The at least one support element may comprise one or more rods or plate structures, each having an end or edge adjacent to or contacting the transcutaneous device for preventing displacement of the transcutaneous device in relation to the carrier part and/or the stationary part in the insertion direction prior to use.

(29) In the initial position of the carrier part, the drive unit may be unbiased FIGS. 1 and 2 show exploded views of an exemplary inserter system of the present invention and FIG. 3 illustrates a cross section of the inserter system 1 along the line B in FIG. 4. The inserter system 1 comprises a

housing part **2** having a sidewall **2'** with a first opening **2A** at a first end and a second opening **2B** at a second end. The inserter system **1** optionally comprises a lid part **4** covering the first opening **2A**. Further, the inserter system comprises a carrier part **6** in an initial position in the housing **4** and a drive unit **8** comprising a first drive part attached to the housing. The carrier part **6** and the drive unit **8** form a first unit **9**. The carrier part **6** comprises a handle portion **7** for enabling a user to manually load the inserter system by moving the carrier part in the retraction direction opposite the insertion direction **14**. The first drive part comprises a first primary drive part **10** and a first secondary drive part **10'**. Each first drive part **10, 10'** comprises assembly means including an assembly recess **11, 11'**, respectively, for engaging with corresponding assembly means formed as assembly protrusions in the housing part **2** for mounting the first unit **9** in the housing part **2**. The drive unit **8** comprises at least one spring element including two first spring elements in the form of leaf spring elements **12** which at their first ends **12A** are connected to the first primary drive part **10** and at their second ends **12B** are connected to the carrier part **6**. Further, the drive unit **8** comprises two second spring elements in the form of leaf spring elements **13** which at their first ends **13A** are connected to the first secondary drive part **10'** and at their second ends **13B** are connected to the carrier part **6**. A configuration with a single spring element such as a helical spring element optionally in combination with guiding means for the carrier part is contemplated. The drive unit **8** is configured for moving the carrier part **6** from a first position to a second position in relation to the housing in an insertion direction **14** along the first axis. The inserter system **1** comprises a transcuteaneous device **16** having a base **17** with a proximal surface **18** and a distal surface **20**, wherein the transcuteaneous device **16** is detachably attached to the carrier part **6** comprising a needle hub **24** including a hub base **26** and optionally an insertion needle **28**. The transcuteaneous device **16** comprises a transcuteaneous element, such as a cannula **34** and/or a sensor element, mounted to the base **17** extending from the proximal surface facing the skin of a patient. The transcuteaneous device may comprise a mounting pad **35** secured to the base **17** and having an adhesive proximal surface for securing the transcuteaneous device to a base plate and/or on the skin of the patient. A removable protective layer or sheet **37** covers the adhesive surface.

(30) The inserter system **1** comprises at least one transport protection element in the form of a first bridge **22** connecting the first primary drive part **10** and the carrier part **6** in the initial position. Further, the inserter system **1** comprises a second bridge **22'** connecting the first secondary drive part **10'** and the carrier part **6** in the initial position. A user loading the inserter system breaks the transport protection elements, i.e. bridges **22, 22'**, when moving the carrier part from the initial position to the first position in relation to the housing. The bridges **22, 22'** are molded as a part of the single unit **9** made of POM. The bridges **22, 22'** each comprise a weakened section having a minimum width of about 1.0 mm and a minimum thickness of about 0.5 mm for having desired strength to withstand transport shocks and at the same time enabling a user to break the bridges when loading the inserter system.

(31) The housing part **2** comprises a first projecting element **40** and a second projecting element **42** extending from the sidewall **2'** and forming a support surface for engagement with locking members **50, 52** of the carrier part **6** for releasably locking the carrier part **6** with the transcuteaneous device **16** in the first position.

(32) The at least one transport protection element comprises at least one support element including a first support element **60** formed in the lid part **4** of the inserter system **1**. The first support element **60** is formed as a cylindrical tube having a first end **62** and a circular cross section with a first diameter  $d_{sub.1}$ , the cylindrical tube extending perpendicular to the first axis. The first support element **60** is arranged to support the transcuteaneous device **16** in the initial position by the first end **62** forming a stop member in the insertion direction. The distance between the first end **62** of the first support element **60** and the protective sheet **37** of the transcuteaneous device should be small enough, such as less than 2.0 mm, to prevent displacement of the transcuteaneous device **16** in relation to the carrier part **6** in the insertion direction prior to use, e.g. during storage or

transportation. In one or more embodiments, the first end **62** contacts the protective sheet **37**. The at least one support element may comprise one or more rods or plate structures, each having an end or edge adjacent to or contacting the transcutaneous device.

(33) The inserter system **1** optionally comprises a packing film **38** that is shrink-fitted around the housing part and the lid part **4**. The packing film **38** is used for showing if the inserter system has been tampered with and therefore may not be sterile.

(34) FIGS. **5** and **6** show different perspective views of the first unit **9** comprising the carrier part **6** and the drive unit **8**. The carrier part **6** comprises a first primary locking element **50** and a second locking element **52** for releasably locking the carrier part **6** in the first position by supporting on the projecting elements **40**, **42**. Alternatively, or in combination, first locking members may be configured to support on first drive part(s) of the drive unit for releasably locking the carrier part in the first position. The locking elements are released by manual deformation of the housing part **2**.

(35) FIGS. **7-9** show an inserter system **101** without a transcutaneous device. The inserter system **101** comprises a locking member **54** detachably mounted in the housing part **2**. The locking member **54** comprises a first part **56** extending along the first axis X and in the initial position engaging with the handle portion **7** of the carrier part **6**. The handle portion **7** forms a cavity, e.g. with rectangular cross section perpendicular to the first axis, for accommodating and supporting the first part **56**. Further, the locking member **54** comprises at least one second locking element including a second primary locking element **58** and a second secondary locking element **59** extending perpendicularly to the first axis and supporting on the stationary part (projecting elements **40**, **42**) of the inserter system **101**. Thereby, the carrier part **6** is prevented from moving in the retraction direction in the initial position such that the cover sheet **30** is not damaged by the carrier part moving due to bumps and shocks during transport. In one or more embodiments, the second locking element(s) **58**, **59** may be arranged for preventing the carrier part from moving in the insertion direction in the initial position, e.g. by extending into one or more recesses in the stationary part. The locking member **54** may comprise a third locking element (not shown) configured for engaging and locking the carrier part to the locking member **54**. The locking member **54** is released by manual rotation, e.g. 90°, of the locking member around the first axis.

(36) FIG. **10** is a cross sectional view of the locking member **54**. The locking member **54** comprises at least one holding element **64**, **64'** for supporting the locking member in the initial position by extending into the cavity of the handle portion **7** preventing unintentional rotation of the locking member **54** when the inserter system is assembled. When the locking member **54** is rotated by a user, the carrier part **6** is forced slightly in the insertion direction by the holding elements **64**, **64'**. The locking member **54** is prevented from rotating in the locked initial position, when the lid part **4** with the first support element **60** is assembled with the housing part **2**.

#### LIST OF REFERENCES

(37) **1**, **101** Inserter system **2** Housing part **2'** Sidewall **2A** First opening **2B** Second opening **4** Lid part **6** Carrier part **7** Handle portion **8** Drive unit **9** First unit **10** First primary drive part **10'** First secondary drive part **12** First spring element **12A** First end **12B** Second end **13** Second spring element **13A** First end **13B** Second end **14** Insertion direction **16** Transcutaneous device **17** Base **18** Proximal surface **20** Distal surface **22** First bridge **22'** Second bridge **24** Needle hub **26** Hub base **28** Insertion needle **30** Seal/cover sheet **34** Cannula **35** Mounting pad **36** Needle protector **37** Protective sheet **38** Packing film **40** First projecting element **42** Second projecting element **50** First primary locking element **52** First secondary locking element **54** Locking member **56** First part **58** Second primary locking element **59** Second secondary locking element **60** First support element **62** First end **64**, **64'** Holding element X First axis

#### Claims

1. An inserter system comprising: a housing; a carrier movably mounted in the housing, wherein the carrier is movable relative to the housing in a first axial direction from an initial position to a retracted position, and in a second axial direction from the retracted position to an insertion position, the first axial direction opposite the second axial direction; a driver mounted in the housing, wherein the driver is operable to move the carrier from the retracted position to the insertion position; a transcutaneous device attached to the carrier; and a breakable bridge engaged between the carrier and a portion of the driver when the carrier is in the initial position to thereby support the carrier in the initial position, wherein the breakable bridge is configured to break in response to movement of the carrier from the initial position to the retracted position.
  2. The inserter system of claim 1, further comprising at least one locking member configured to selectively limit movement of the transcutaneous device.
  3. The inserter system of claim 2, wherein the at least one locking member comprises a first locking member and a second locking member operable to engage the first locking member; wherein the housing comprises the first locking member; and wherein the carrier comprises the second locking member.
  4. The inserter system of claim 1, further comprising a gas permeable seal or cover sheet.
  5. The inserter system of claim 1, wherein the transcutaneous device is detachably attached to the carrier part.
  6. An inserter system comprising: a housing having an open proximal end; a carrier movably mounted in the housing, wherein the carrier is movable from an initial position to a retracted position, and from the retracted position to an insertion position; a transcutaneous device attached to the carrier, wherein movement of the carrier from the retracted position to the insertion position moves a portion of the transcutaneous device beyond the proximal end of the housing; a gas permeable seal or cover sheet covering the open proximal end of the housing; and a locking member configured to selectively retain the carrier in the initial position to thereby discourage damaging of the gas permeable seal or cover sheet by the carrier and/or the transcutaneous device.
  7. The inserter system of claim 6, further comprising a driver mounted in the housing, wherein the driver is operable to move the carrier from the retracted position to the insertion position.
  8. The inserter system of claim 7, further comprising a breakable bridge positioned between the carrier and a portion of the driver when the carrier is in the initial position to thereby support the carrier in the initial position.
  9. The inserter system of claim 6, wherein the gas permeable seal or cover sheet allows for sterilization of the inserter system.
  10. The inserter system of claim 6, further comprising a support configured to limit movement of the transcutaneous device relative to the carrier.
-